BDA Course 2026

Lecture 9

Objectives of this Lecture

- Model comparison with LOO-CV
- When is cross-validation applicable?
- Hypothesis testing

Recap previous lecture

- Posterior (prior) predictive checks
- Cross-validation
  - Fast cross-validation with PSIS-LOO-CV

The log-score

- We want to know how well the model predicts new data

- Log score: $\log p(\tilde{y} \mid y, M)$
  - theoretically justified
  - uses the whole predictive distribution

- Application specific cost/utility functions can be better, and metrics like R^2, RMSE, etc can be easier to interpret

Expected log predictive density

- Log predictive density of the observed data (training fit)

$$\text{lpd} = \sum_{i=1}^{n} \log p(y_i \mid y)$$

  y_i is used both for fitting and for evaluation, so lpd is optimistic

- use LOO-CV to approximate the elpd

$$\text{elpd_loo} = \sum_{i=1}^{n} \log p(y_i \mid y_{-i})$$

- $\text{lpd} - \text{elpd_loo} = p_{\text{loo}}$ effective number of parameters (under certain conditions)

Fast LOO with PSIS

- Brute-force LOO needs n MCMC runs
- We can use importance sampling and only one fit
  - draws from the full posterior $\theta^{(s)} \sim p(\theta \mid x, y)$ are the proposal
  - the LOO posterior $p(\theta \mid x_{-i}, y_{-i})$ is the target

$$w_i^{(s)} \propto \frac{1}{p(y_i \mid x_i, \theta^{(s)})}$$

- Raw weights can have very high or infinite variance
- Pareto smoothing: fit a generalized Pareto distribution to the tail (largest weights) and replace them with smoothed values

Recap: what does Pareto-$\hat{k}$ tell us?

- $\hat{k}$ is the shape of the fitted Pareto tail
  - the weights have only $\frac{1}{\hat{k}}$ finite moments (e.g. $\hat{k} < 0.5$ means finite variance)
  - it tells how far the LOO posterior is from the full posterior
- Rule of thumb
  - $\hat{k} < 0.7$: PSIS estimate is reliable
  - $\hat{k} > 0.7$: importance sampling is unreliable for that observation
- High $\hat{k}$ means observation i is highly influential
  - possible outlier, data error or model misspecification
- If there are many high values
  - moment matching, `reloo`, integrated LOO or K-fold-CV
  - `p_loo` can help as a diagnostic

Recap: PSIS-LOO is more than elpd

- We can use the PSIS weights to approximate the LOO posterior, then any LOO quantity

$$E_{\theta \mid x_{-i}, y_{-i}}[h(\theta)] \approx \sum_{s=1}^{S} \tilde{w}_i^{(s)} h\left(\theta^{(s)}\right)$$

- Examples
  - LOO predictive means, quantiles and event probabilities
  - MAE, RMSE, classification accuracy, CRPS
  - LOO-R^2
  - LOO-PIT for calibration checking

Model comparison

- Often we end-up having multiple models for the same data
- We can use posterior predictive checks to discard/improve poorly fitting models
- Often we reach a point where multiple models seem reasonable
- The predictive performance can be compared using measures like elpd_loo

Using the ELPD for model comparison

- The model with the higher elpd is estimated to have better predictive performance
- Given a model M_A and M_B

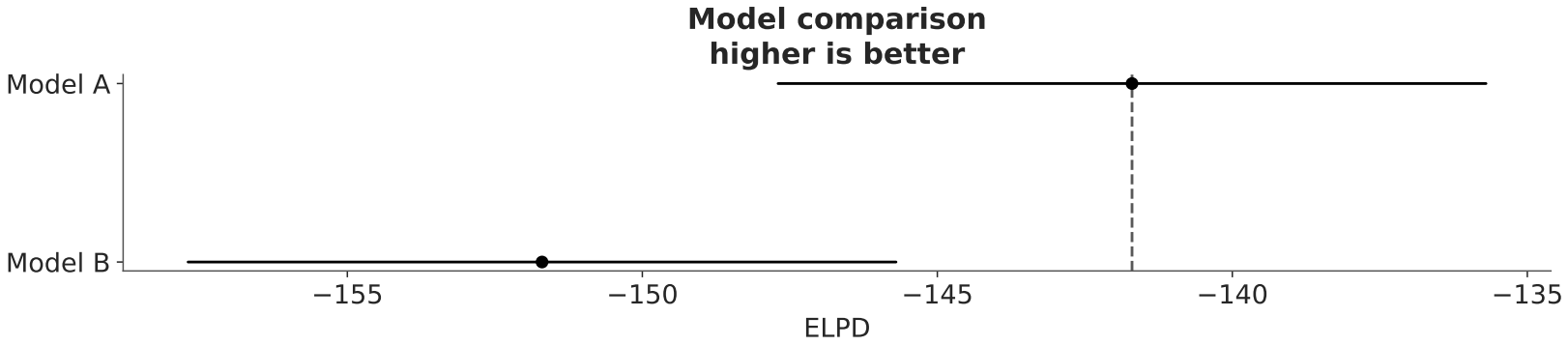
$$\text{elpd_diff} = \text{elpd}_A - \text{elpd}_B$$

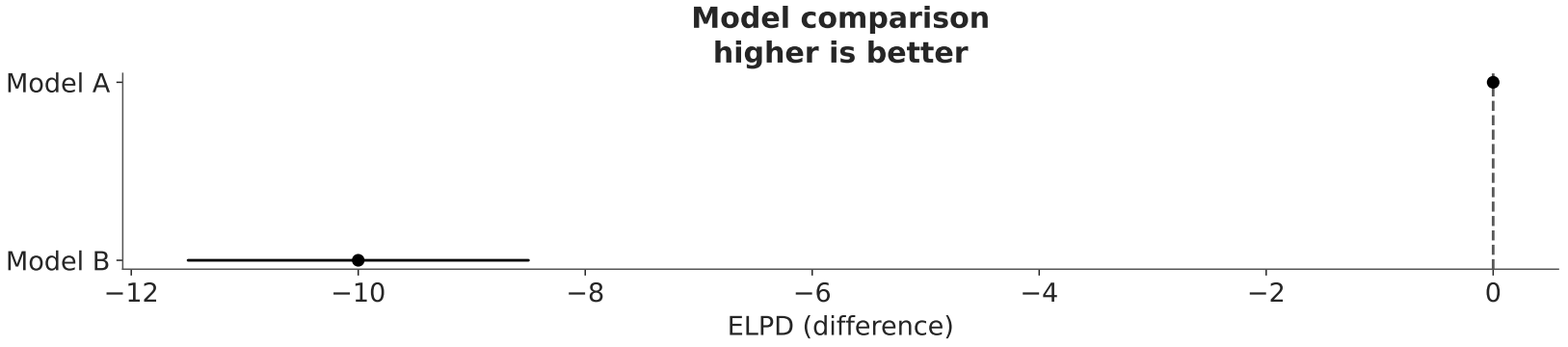
  - $\text{elpd_diff} > 0$: M_A is estimated to predict new data better

- As we already stablish PSIS-LOO-CV, give as a fast and reliable approximation to the elpd_loo
- elpd_loo, and their difference, are point estimates, we also need to consider the uncertainty

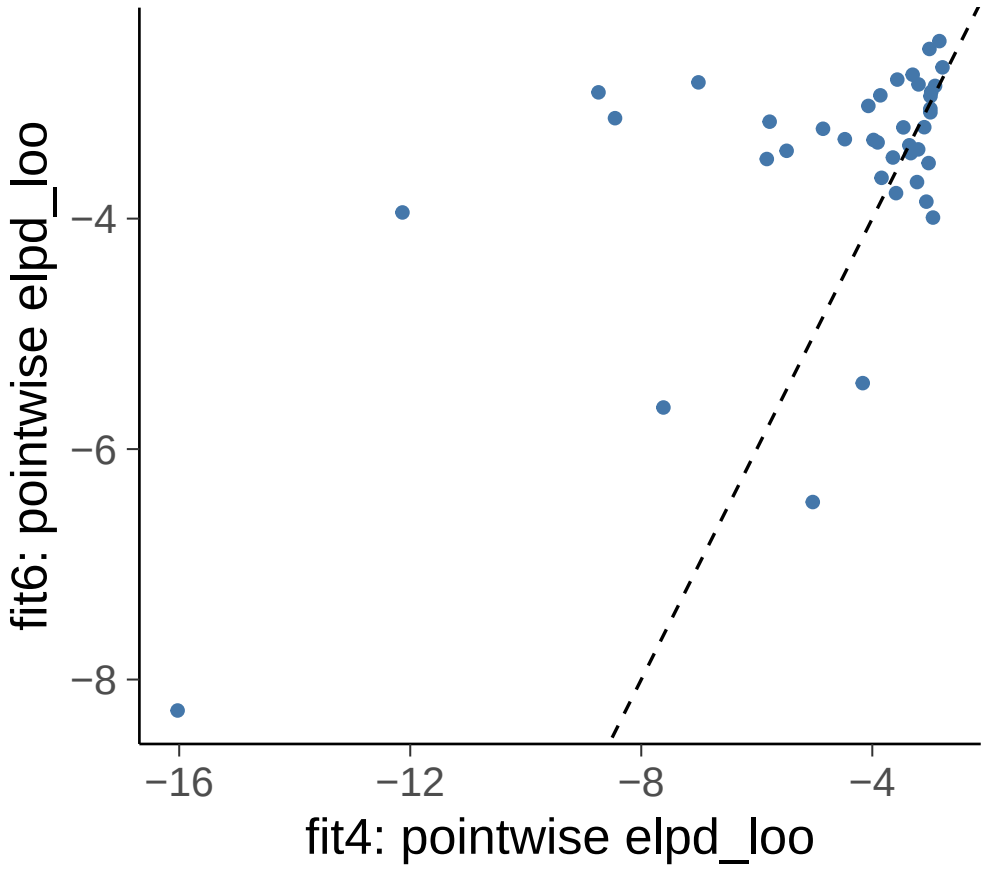
SE for elpd_loo and their differences

- We should compare the elpd_loo difference and the SE of the difference





SE for elpd differences



Latent hierarchical linear (fit4) vs latent hierarchical linear + spline (fit6)

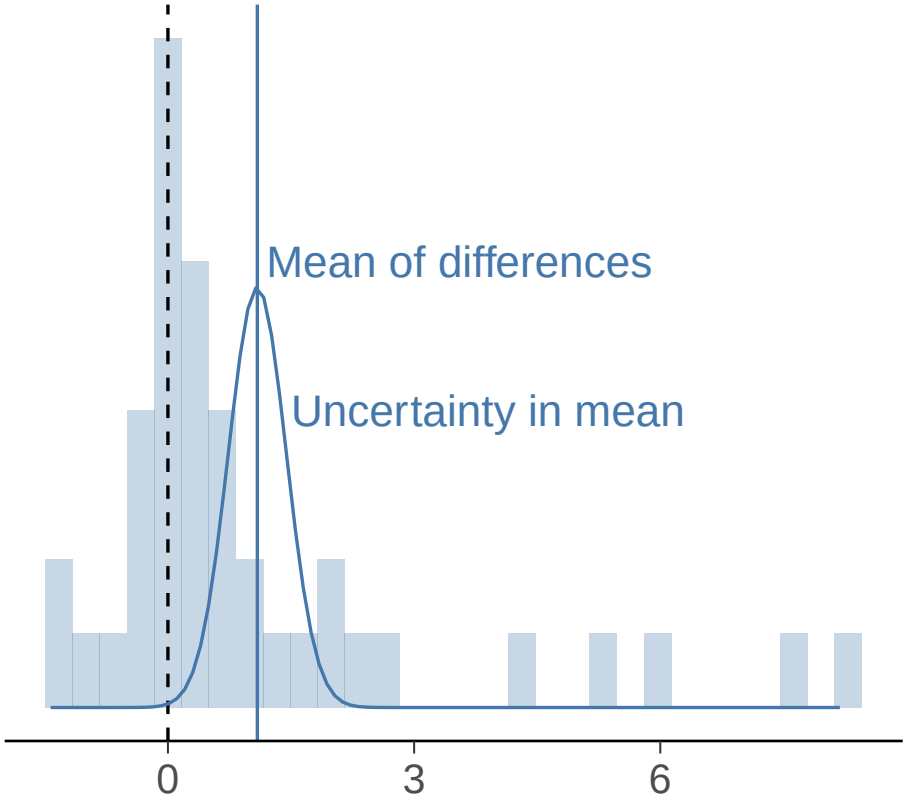
SE of the difference

- Both elpd_loo estimates are computed from the same n observations
  - pointwise terms $\log p(y_i \mid y_{-i})$ are correlated between the models
- Comparing the two SEs separately ignores this correlation
  - $\mathrm{Var}(A - B) = \mathrm{Var}(A) + \mathrm{Var}(B) - 2\ \mathrm{Cov}(A, B)$
- Instead, compute the pointwise difference $d_i = \log p(y_i \mid y_{-i}, M_A) - \log p(y_i \mid y_{-i}, M_B)$

$$\text{elpd_diff} = \sum_{i=1}^{n} d_i, \quad \text{SE}_{\text{diff}} = \text{sd}(d_i) \cdot \sqrt{n}$$

We are assuming the differences d_i are normally distributed

SE for elpd differences



fit6 vs fit4: pointwise elpd_loo difference

- To compare two models we compute the pointwise difference of the ELPDs and we use a normal approximation to estimate its SE

Reliability of the SE for elpd differences

- The normal approximation works well when $|\text{elpd_diff}| > 4$, the models are well specified and $n > 100$

- $|\text{elpd_diff}| < 4$: the SE is not reliable, but the difference is small anyway
- The models are badly misspecified, with outliers in the data
  - check and extend the model first
- The number of observations is small
  - any inference with small n is difficult

Frequently asked questions on CV.

Sivula, Magnusson et al. 2025. Uncertainty in Bayesian leave-one-out cross-validation based model comparison. *Bayesian Analysis*, doi:10.1214/25-BA1569.

SE for elpd_loo and their differences

- We should compare the elpd_loo difference and the SE of the difference

Latent hierarchical linear + spline

```
> loo(fit6)
         Estimate   SE
elpd_loo   -141.7  7.2
p_loo        10.9  2.5
```

Latent hierarchical linear

```
> loo(fit4)
         Estimate   SE
elpd_loo   -184.3 17.3
p_loo        23.8  5.7
```

```
> loo_compare(loo(fit4), loo(fit6))
     elpd_diff se_diff
fit6   0.0       0.0
fit4 -42.6      14.3
```

Log score and elpd_loo

- Log score is not easily interpretable
- but is information theoretically good utility for the goodness of the whole distribution
- and thus is useful in model comparison

Log score and elpd_loo

- Interpretation in discrete case
  - log probability
- For example, in student retention
  - $\frac{1}{N}\sum_{n=1}^{N}\exp\left(\text{elpd}_{\text{loo},n}\right) \approx 4\%$ probability that we predict the observed value
  - compare to guessing uniformly from the data range [121,310] having $\frac{1}{310-121+1} \approx 0.5\%$ probability
- Interpretation in continuous case
  - can be compared to a simple reference distribution

What do we want to predict?

- LOO leaves out one observation. But, is that always the right question?
- Imagine a model for the weights of rats, each rat measured several weeks in a row
- Which prediction do we care about?
  1. another measurement of a rat that is already in the data
  2. a new rat that is not in the data
  3. the same rats next month
- These are three different prediction tasks, so three different cross-validations

Hierarchical data: existing or new group?

- Prediction for an existing group
  - leave out one observation: LOO
  - the group's own parameters are still learned from its other observations
- Prediction for a new group
  - leave-one-group-out (LOGO)
  - predictions for new groups can only rely on the population level
- LOO is often a good first check, since it can already reveal problems with the model for the observations

Riha et al., 2026. Approximating Bayesian leave-one-group-out cross-validation https://arxiv.org/abs/2609.05713

Time series: forecasting and LOO

- Usually we want to predict the future
  - future observations cannot be used to predict the past
  - Leave-future-out (LFO)
- LOO predicts y_t using observations before and after t
  - it tells us how well each observation is predicted from the others (noise distribution, outliers, abrupt changes)
  - cheap with PSIS-LOO-CV, so a good first check
  - also matches the task if we want to fill in missing values in the middle of the series (e.g. a sensor failure)

Bürkner, Gabry & Vehtari, 2020. Approximate leave-future-out cross-validation for Bayesian time series models. https://www.tandfonline.com/doi/full/10.1080/00949655.2020.1783262

Summary: data-generating mechanisms and prediction tasks

- Cross-validation can be used to evaluate different prediction tasks
  - Choose a validation strategy that reflects the intended prediction task
  - Use knowledge of the data-generating process and practical objectives whenever available
- Cross-validation can be useful even when there is no clearly defined prediction task
  - It can help assess predictive performance and compare models

see Vehtari & Ojanen (2012) and CV-FAQ

Pareto smoothed importance sampling CV variants

- PSIS-LOO for hierarchical models
  - leave-one-group out is challenging for PSIS-LOO
  - Stan demo of the challenges and integrated LOO at roaches
  - see also Merkel, Furr and Rabe-Hesketh (2018)
- PSIS-LOO for non-factorized models
  - mc-stan.org/loo/articles/loo2-non-factorizable.html
- PSIS-LOO for time series
  - Approximate leave-future-out cross-validation (LFO-CV)
  - mc-stan.org/loo/articles/loo2-lfo.html

K-fold cross-validation

- K-fold cross-validation can approximate LOO
  - the same use cases as with LOO
- K-fold cross-validation can be used for hierarchical models
  - good for leave-one-group-out
- K-fold cross-validation can be used for time series
  - with leave-block-out

When do we want predictive model selection?

- Models used for making predictions: predictive model selection is the natural choice
- Parameter of interest: look at its posterior directly, often easier and more accurate
  - instead of comparing
    Model 1: $y \sim \text{normal}(\alpha, \sigma)$
    vs
    Model 2: $y \sim \text{normal}(\alpha + \beta x, \sigma)$
    look at the posterior of β
- No directly interpretable parameter (e.g. spline-based terms): predictive model selection is useful even if prediction is not the goal

Variable selection

- Variable selection aims to identify the most relevant predictors
- We can see it as a special case of model comparison for many model
  - With p predictors there are 2^p possible models
- In principle we can use CV (and PSIS-LOO-CV) to compare them
- In practice with many (potentially similar) models overfitting is likely
- We will discuss an alternative approach: projection predictive variable selection

Model averaging

- Bayesian Model averaging (BMA):
  - combines predictions from multiple models weighted by their posterior model probabilities
- Bayesian stacking
  - chooses the weights by optimizing the predictive performance of the combination
  - may work better than BMA when the models are misspecified or small data
- Oftern we prefer model expansion
  - hierarchical models
  - e.g. Student-t instead of normal (normal is the limit when $\nu \to \infty$)
  - nested models: expand so that the simpler model is a special case

Yao, Vehtari, Simpson, and Gelman (2018). Using stacking to average Bayesian predictive distributions (with discussion). *Bayesian Analysis*, 13(3):917-1003

looic?

```
> loo(fit6)

Computed from 4000 by 40 log-likelihood matrix

         Estimate   SE
elpd_loo   -141.7  7.2
p_loo        10.9  2.5
looic       283.4 14.4
------
Monte Carlo SE of elpd_loo is 0.1.
```

- `loo` output shows also looic for historical reasons
  - it's −2 * elpd_loo
  - ignore it, it will be removed in the future

Information criteria

Information criteria estimate predictive performance, too

- AIC uses maximum likelihood estimate for prediction
- DIC uses posterior mean for prediction
- ...
- WAIC is the only Bayesian information criterion

Vehtari & Ojanen (2012). A survey of Bayesian predictive methods for model assessment, selection and comparison. *Statistics Surveys*, 6:142-228.

WAIC vs PSIS-LOO

- WAIC has the same target and assumptions as LOO
- PSIS-LOO is more accurate
- PSIS-LOO has better diagnostics
- We can often “fix” problematic PSIS-LOO cases with moment matching and reloo
- LOO makes the prediction assumption more clear, which helps if K-fold-CV is needed instead

Vehtari, Gelman and Gabry (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5):1413–1432

Vehtari & Ojanen (2012). A survey of Bayesian predictive methods for model assessment, selection and comparison. *Statistics Surveys*, 6:142-228.

Marginal likelihood and Bayes factor

Bayes Factor $\frac{p(y|M_1)}{p(y|M_2)}$

Marginal likelihood $p(y|M_1) = \int p(y|\theta, M_1)p(\theta|M_1)d\theta$

Marginal likelihood with chain rule:
$p(y|M_1) = p(y_1|M_1)p(y_2|y_1, M_1)\cdots p(y_n|y_1, \ldots, y_{n-1}, M_1)$

- Bayes factor is extremely sensitive to the prior
  - the first term is a prediction from the prior alone and the subsequent terms incorporate the observed data sequentially so the result is very sensitive to the prior

Vehtari & Ojanen (2012). A survey of Bayesian predictive methods for model assessment, selection and comparison. *Statistics Surveys*, 6:142-228.

Hypothesis testing

- Is there an effect? A dichotomous question: compare "no effect" (H_0: $\beta = 0$) with "some effect"
- Frequentist: null hypothesis significance testing
  - p-value: probability of data at least as extreme as observed, if H_0 is true
  - reject H_0 if the p-value is below a threshold (e.g. 0.05)
- Bayesian: compare the models with the Bayes factor $\frac{p(y|M_1)}{p(y|M_2)}$
  - the ratio of the marginal likelihoods from the previous slides



Common statistical tests as Bayesian models

- Most common statistical tests are linear models

test	model	formula
t-test	mean of data	`y ~ 1`
paired t-test	mean of diffs	`(y1 - y2) ~ 1`
Pearson correl.	linear model	`y ~ 1 + x`
two-sample t-test	group means	`y ~ 1 + gid`
ANOVA	hier. model	`y ~ 1 + (1 | gid)`

- So often instead of thinking about this “named tests” we can think in terms of the underlying linear models
  - Assumption may be more transparent and easier to check
  - We can extend them to better match our problem

Check out lindeloev.github.io/tests-as-linear and Regression and other stories

Beta blockers

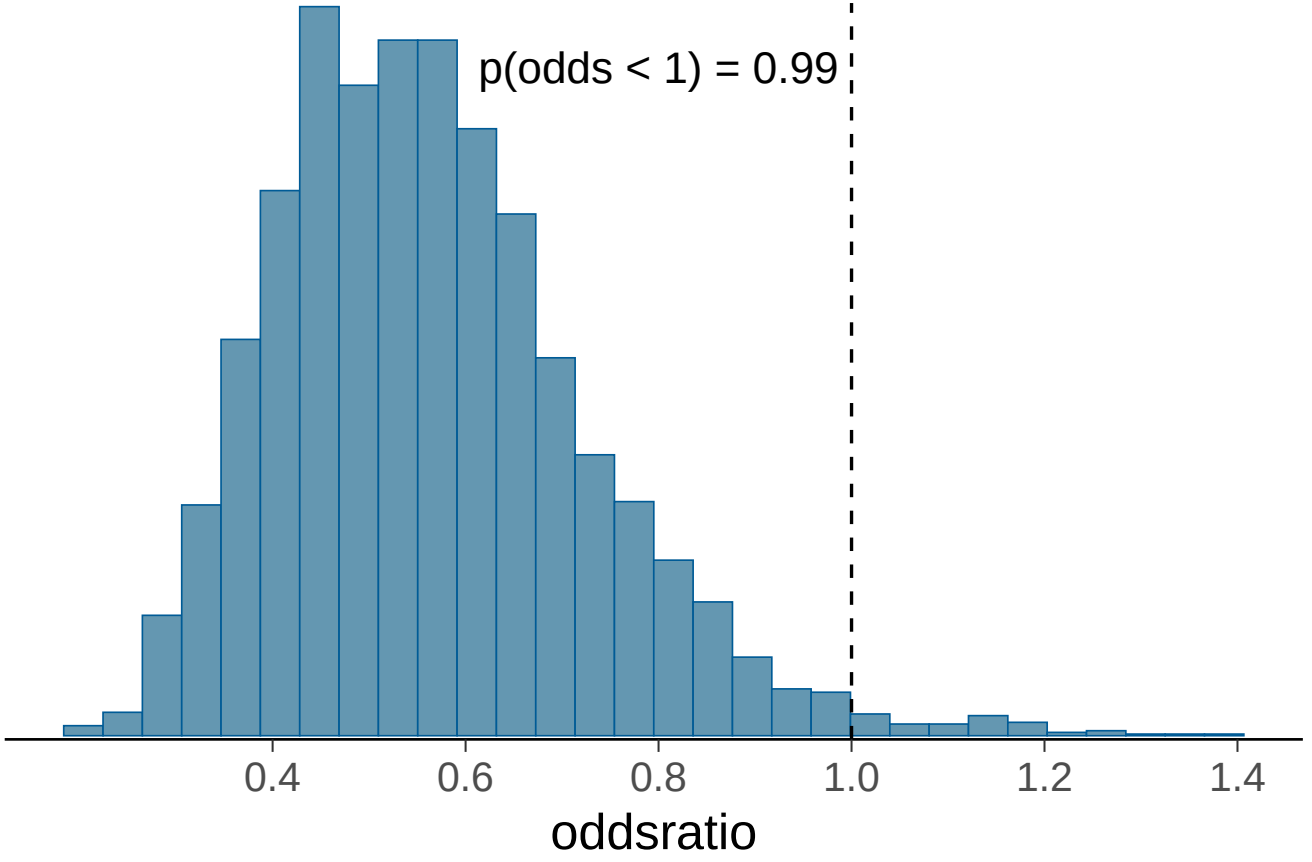
- Does the treatment reduce mortality of cardiac patients?
  - patients *randomly* assigned to *control* and `treatment`
  - out of 674 controls, 39 died; out of 680 treated, 22 died

```
fitb1 <- brm(y | trials(N) ~ 1, family = binomial(), data = d_bin2)
fitb2 <- brm(y | trials(N) ~ 1 + grp2, family = binomial(), data = d_bin2)
```

- elpd difference is small (less than 4) and smaller than its SE: CV does not tell the models apart

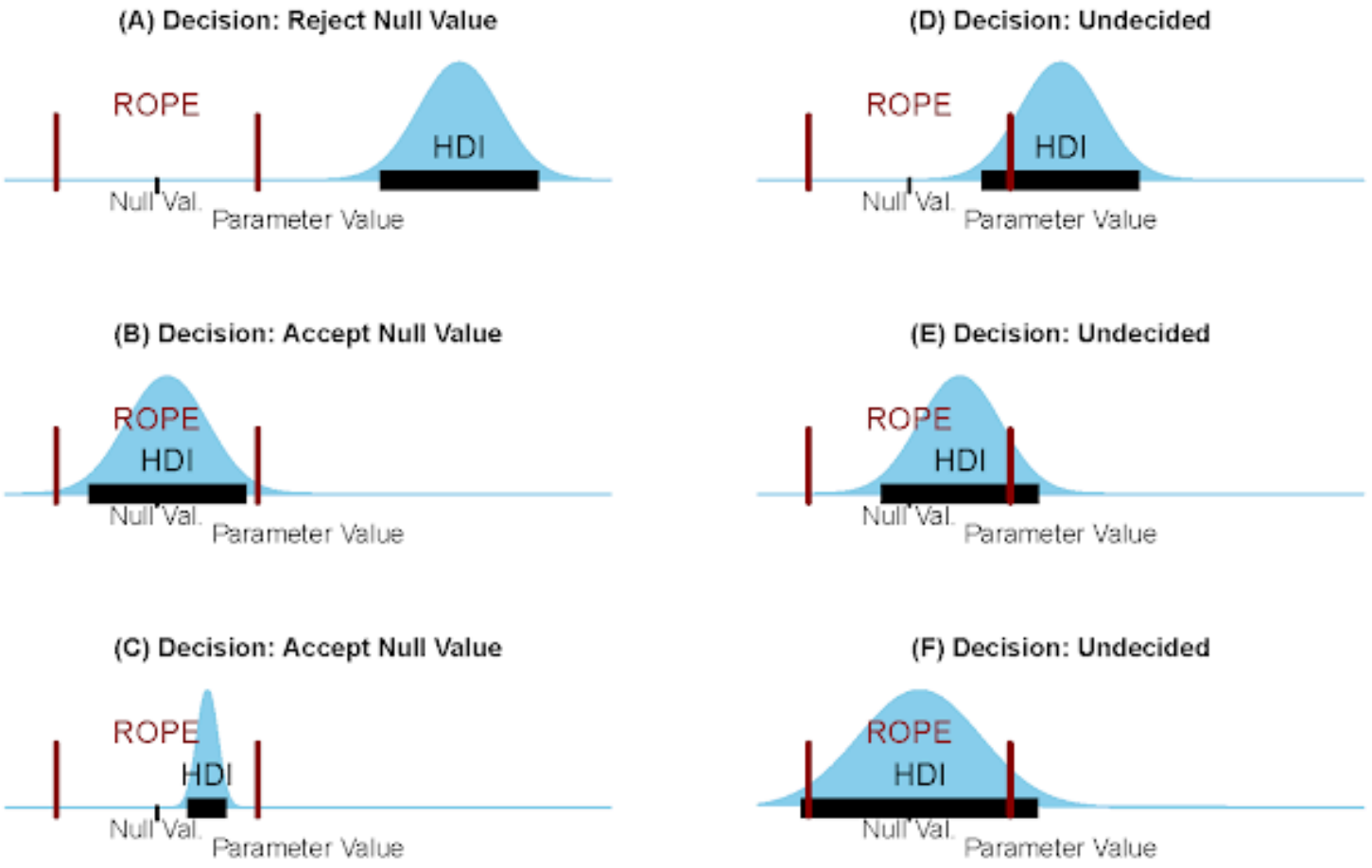
Beta blockers

- The posterior of the treatment effect answers the question more directly: how large is it, and how certain are we of its sign?



Region of practical equivalence (ROPE)

- Define a range of effect sizes that are practically equivalent to no effect
- Compare if a credible interval overlaps with the ROPE



- Instead of a CI we can use all the draws from the posterior

Often we do not need a dichotomous decision

- We can report the full posterior distribution of the effect
  - this allows us to see the range of plausible effect sizes
  - we can quantify the probability that the effect is positive, negative, or practically negligible
  - this approach avoids forcing a binary decision and provides a richer summary of the evidence